

Aldehydes And Ketones

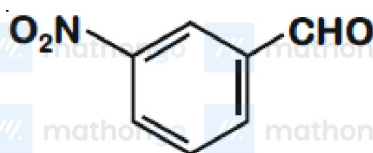
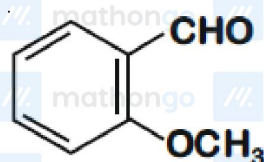
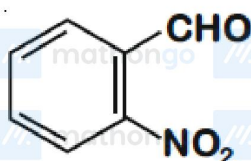
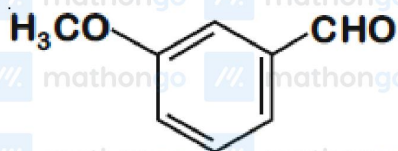
Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

The increasing order of the following compounds towards HCN addition is



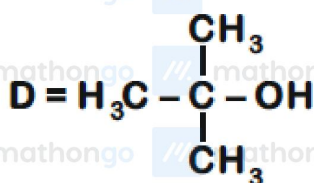
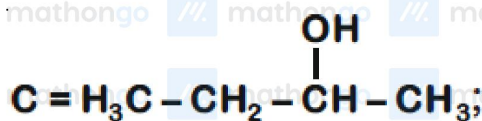
- (A) (iii) < (iv) < (ii) < (i)
(B) (iii) < (i) < (iv) < (ii)
(C) (iii) < (iv) < (i) < (ii)
(D) (i) < (iii) < (iv) < (ii)

Q2 JEE Main 2020 - 2 September (Evening)

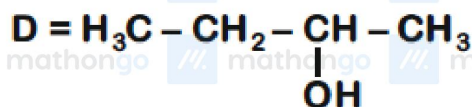
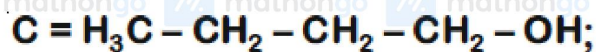
Two compounds *A* and *B* with same molecular formula (C_3H_6O) undergo Grignard's reaction with methylmagnesium bromide to give products *C* and *D*. Products *C* and *D* show following chemical tests.

Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are



(A)



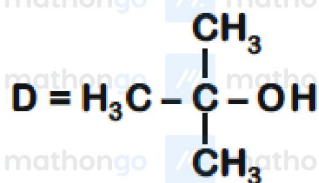
(B)

Aldehydes And Ketones

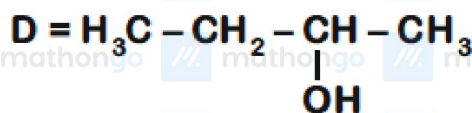
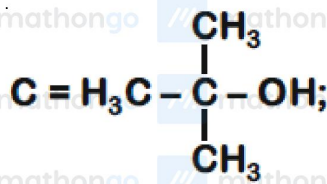
Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry



(C)



(D)

Q3 JEE Main 2020 - 3 September (Evening)

The increasing order of the reactivity of the following compounds in nucleophilic addition reaction is
Propanal, Benzaldehyde, Propanone, Butanone

- (A) Propanal < Propanone < Butanone < Benzaldehyde
- (B) Benzaldehyde < Propanal < Propanone < Butanone
- (C) Benzaldehyde < Butanone < Propanone < Propanal
- (D) Butanone < Propanone < Benzaldehyde < Propanal

Q4 JEE Main 2020 - 3 September (Evening)

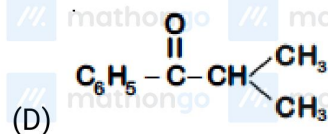
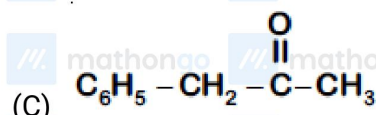
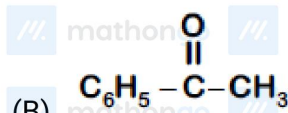
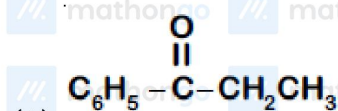
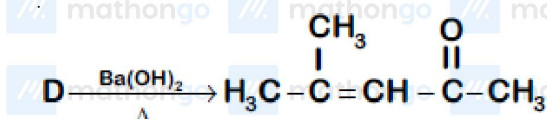
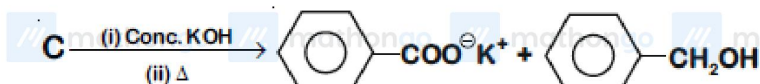
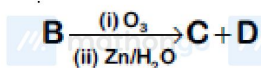
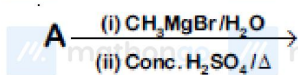
Aldehydes And Ketones

Questions with Answer Keys

JEE Main 2020 Chapterwise

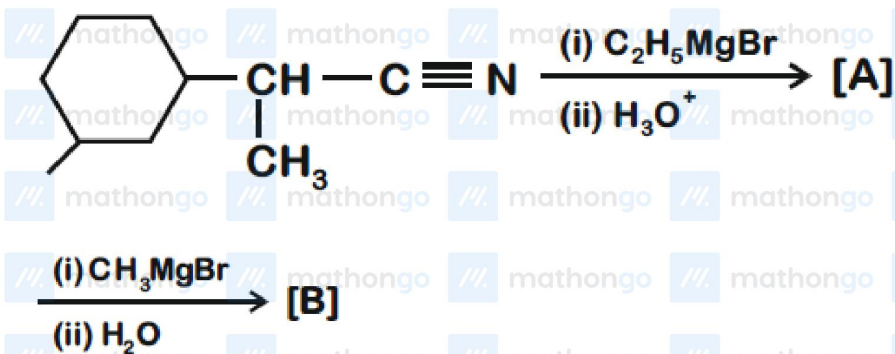
Chemistry

The compound **A** in the following reactions is



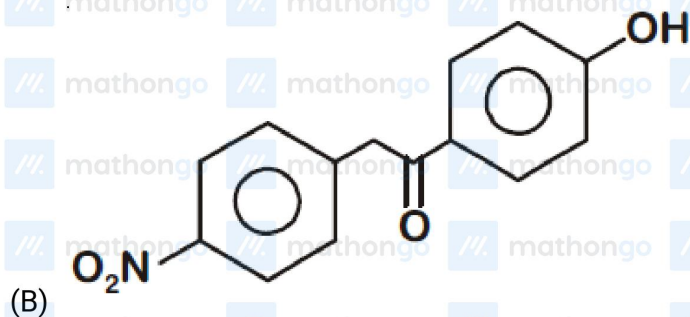
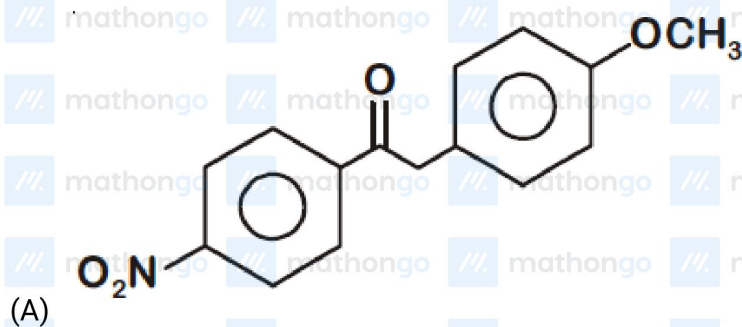
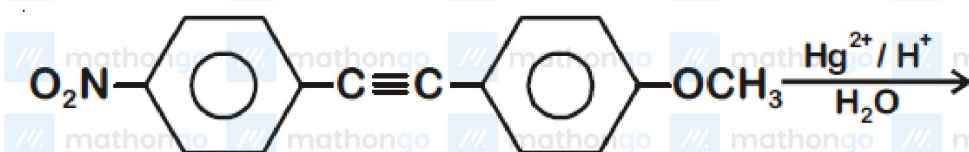
Q5 JEE Main 2020 - 4 September (Morning)

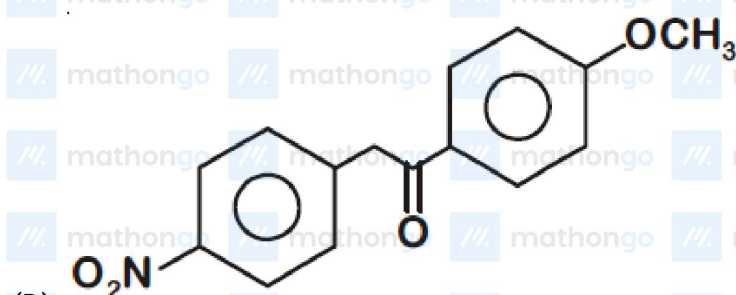
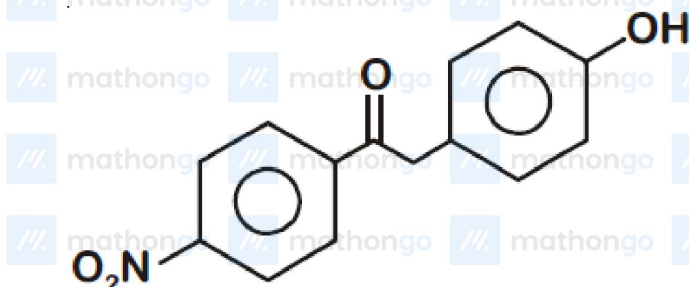
The number of chiral centres present in [B] is



Q6 JEE Main 2020 - 6 September (Morning)

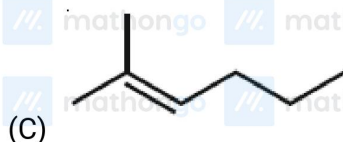
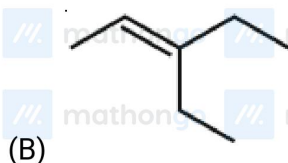
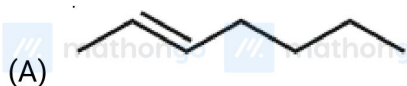
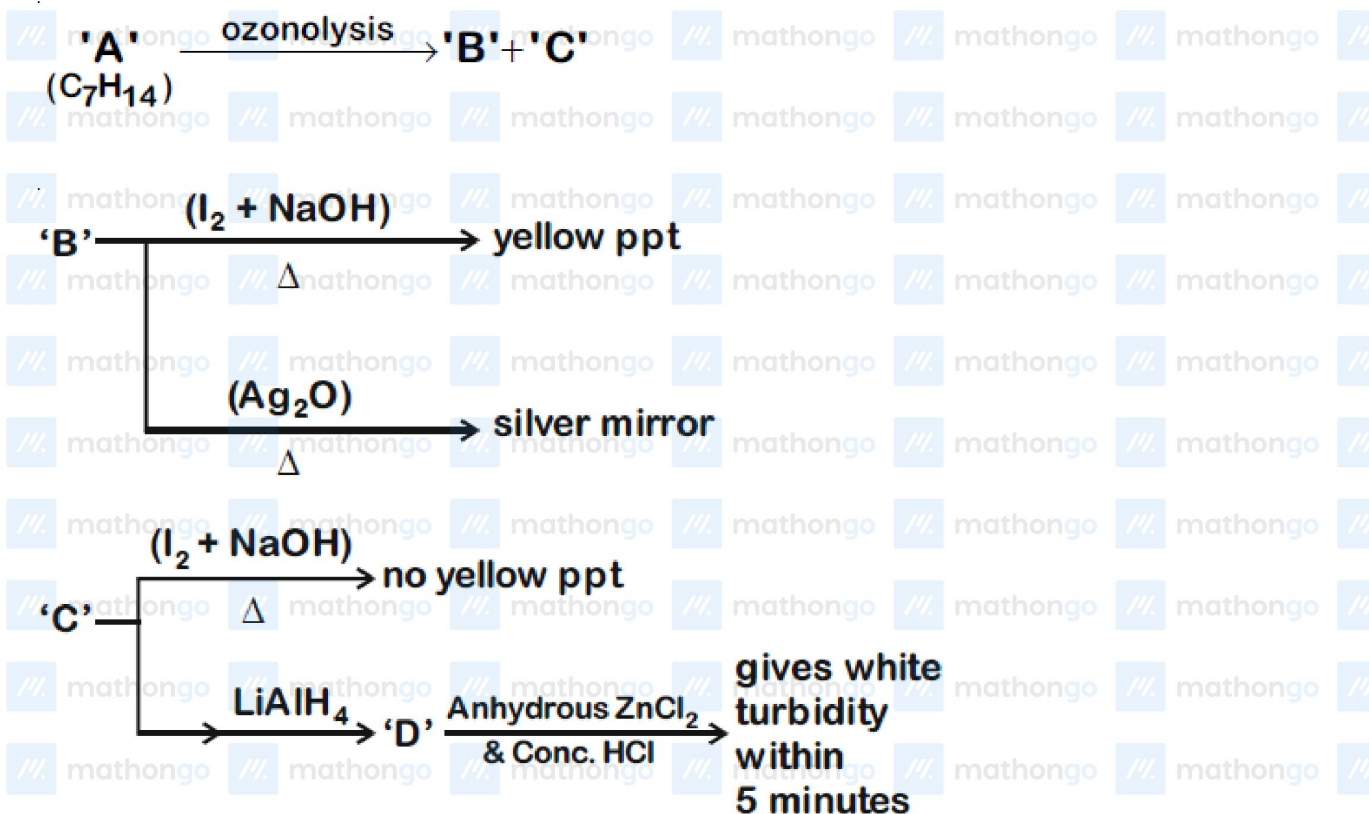
The major product obtained from the following reaction is





Q7 JEE Main 2020 - 6 September (Morning)

Consider the following reactions



Aldehydes And Ketones

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry



(D)

Q8 JEE Main 2020 - 6 September (Evening)

The correct match between Item-I (starting material) and Item-II (reagent) for the preparation of benzaldehyde is

Item-I

Item-II

(I) Benzene

(P) HCl and SnCl_2 ,

H_3O^+

(II) Benzonitrile

(Q) H_2 , Pd-BaSO₄, S

and quinoline

(III) Benzoyl Chloride

(R) CO, HCl and AlCl_3

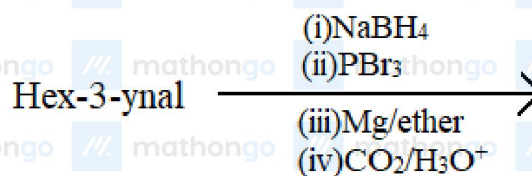
(A) (I) $\rightarrow$ (R), (II) $\rightarrow$ (P) and (III) $\rightarrow$ (Q)

(B) (I) $\rightarrow$ (P), (II) $\rightarrow$ (Q) and (III) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q), (II) $\rightarrow$ (R) and (III) $\rightarrow$ (P)

(D) (I) $\rightarrow$ (R), (II) $\rightarrow$ (Q) and (III) $\rightarrow$ (P)

Q9 JEE Main 2020 - 7 January (Morning)



What is the product of following reaction ?

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(A) 



(B) 



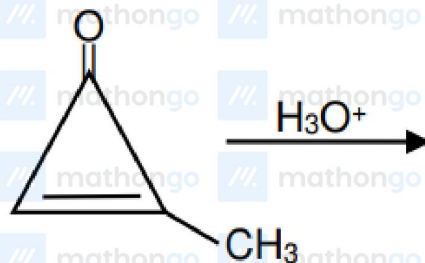
(C) 



(D) 

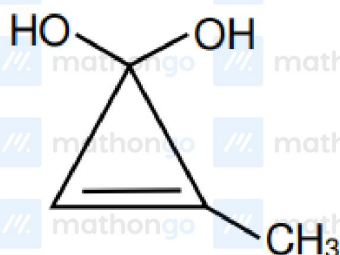


Q10 JEE Main 2020 - 8 January (Evening)

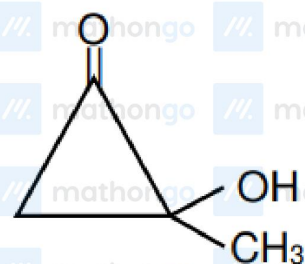


The major product of the following reactions is

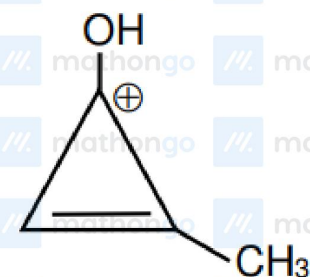
(A)



(B)



(C)



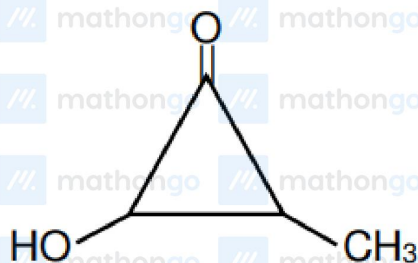
Aldehydes And Ketones

Questions with Answer Keys

JEE Main 2020 Chapterwise

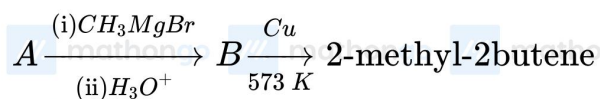
Chemistry

(D)



Q11 JEE Main 2020 - 9 January (Evening)

Consider the following reactions



The mass percentage of carbon in *A* is ...

Answer Key

Q1 (B)

Q2 (A)

Q3 (D)

Q4 (C)

Q5 (4)

Q6 (D)

Q7 (B)

Q8 (A)

Q9 (A)

Q10 (C)

Q11 (66.67)

Q1

Attack of nucleophile on carbonyl centres depends upon

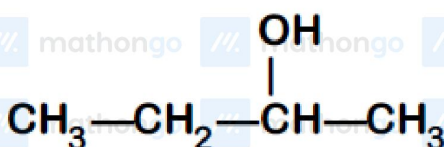
(i) Steric factor

(ii) Electronic factor

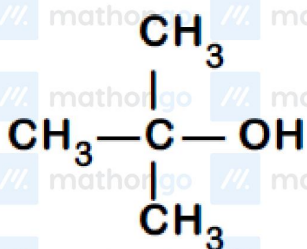
∴ Rate of reaction should follow the order

(ii) > (iv) > (i) > (iii)

Q2



can give positive CAN test, iodoform test, and with Lucas reagent it takes 5 minutes to give turbidity.



can not give iodoform test but can give CAN test and Lucas test.

Q3

Increasing hindrance and electron releasing group (ERG) attached on carboxyl group decrease the reactivity.

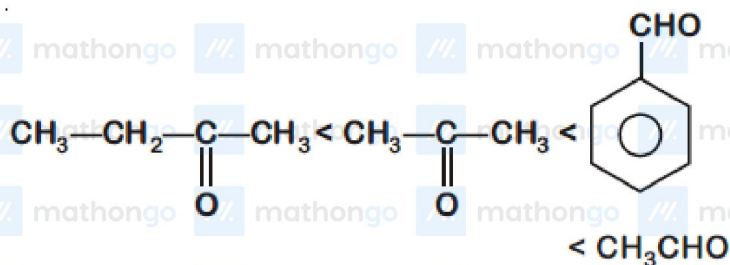
Aldehydes And Ketones

Hints & Solutions

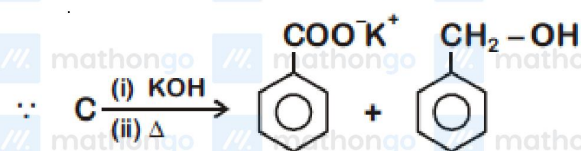
JEE Main 2020 Chapterwise

Chemistry

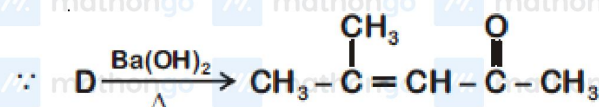
Correct order of nucleophilic addition reaction is



Q4



$\therefore$ C is Ph - CHO

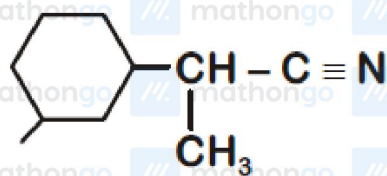


$\therefore$ D is $\text{CH}_3\text{--}\overset{\text{O}}{\text{C}}\text{--CH}_3$

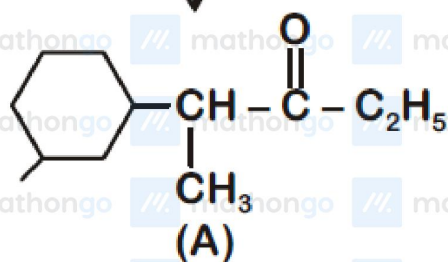
B is $\text{Ph--CH=C(CH}_3\text{)--CH}_3$

A is $\text{C}_6\text{H}_5\text{--CH}_2\text{--}\overset{\text{O}}{\text{C}}\text{--CH}_3$

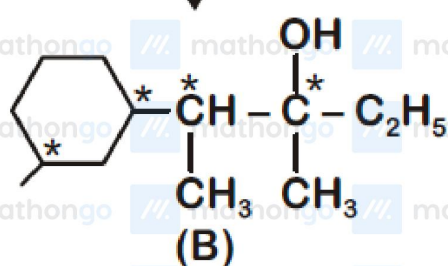
Q5



(i) $\text{C}_2\text{H}_5\text{MgBr}$ (ii) H_3O^+



(i) CH_3MgBr (ii) H_2O



Number of chiral centres in the product (B) = 4.

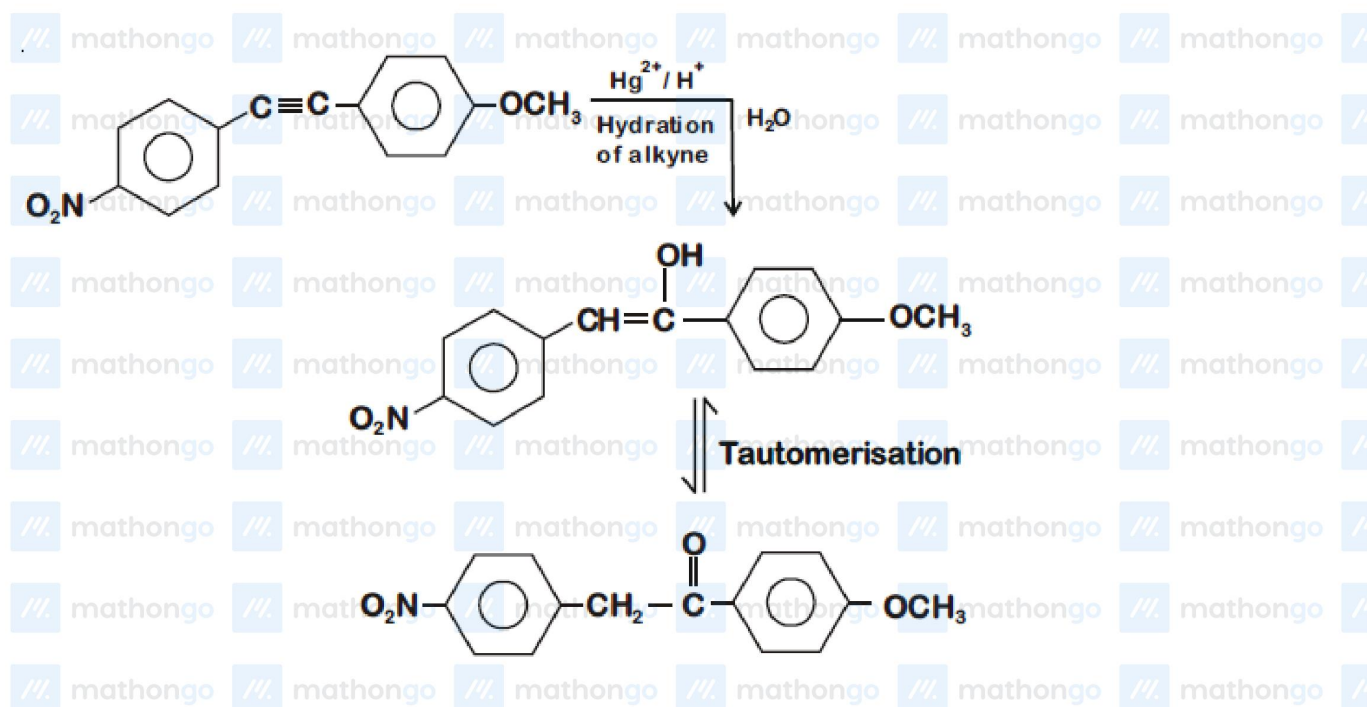
Q6

Aldehydes And Ketones

Hints & Solutions

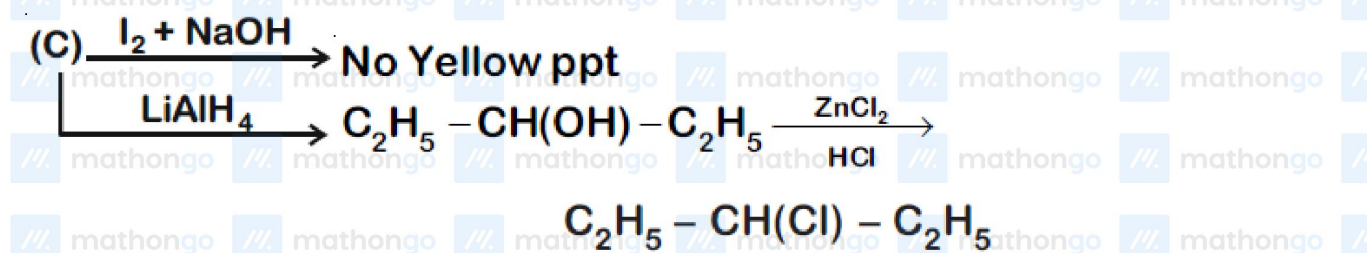
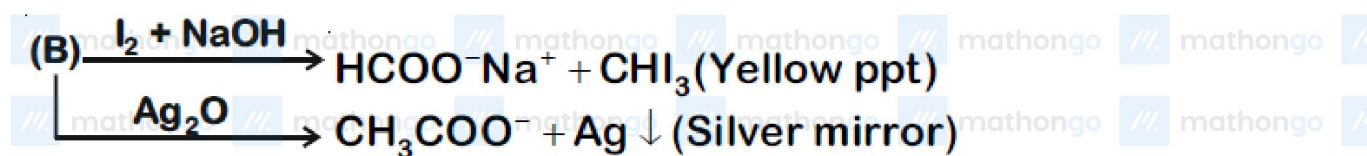
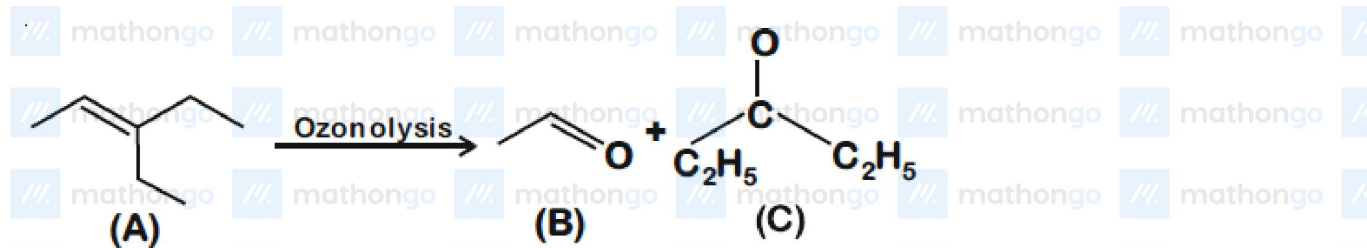
JEE Main 2020 Chapterwise

Chemistry



Ethers are least reactive and their cleavage takes place under drastic conditions.

Q7



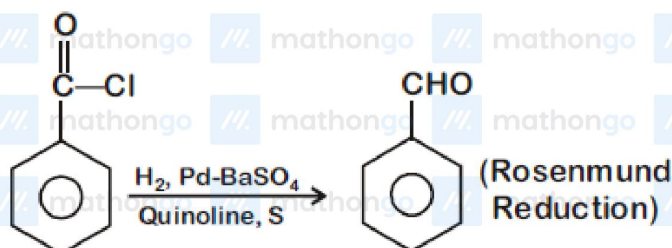
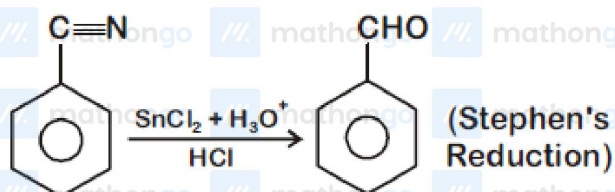
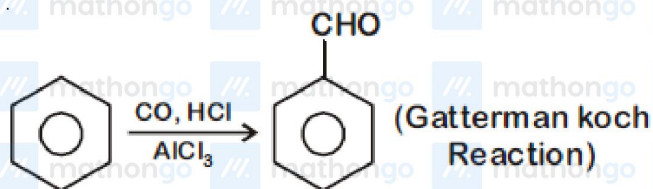
Q8

Aldehydes And Ketones

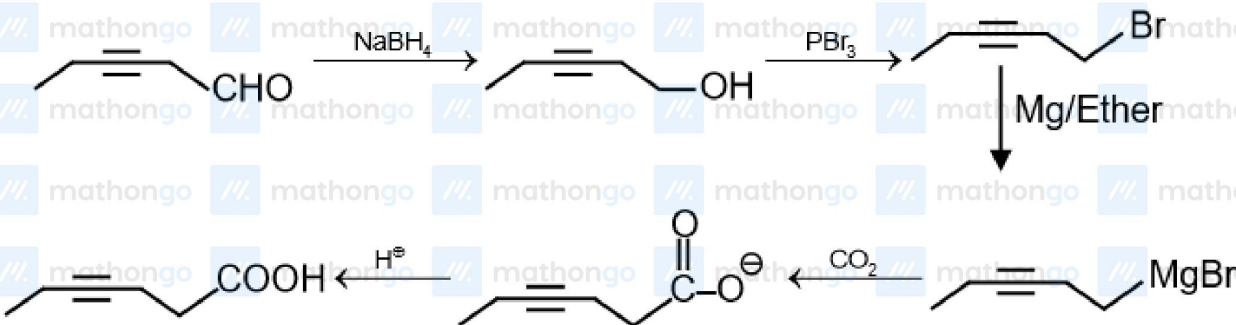
Hints & Solutions

JEE Main 2020 Chapterwise

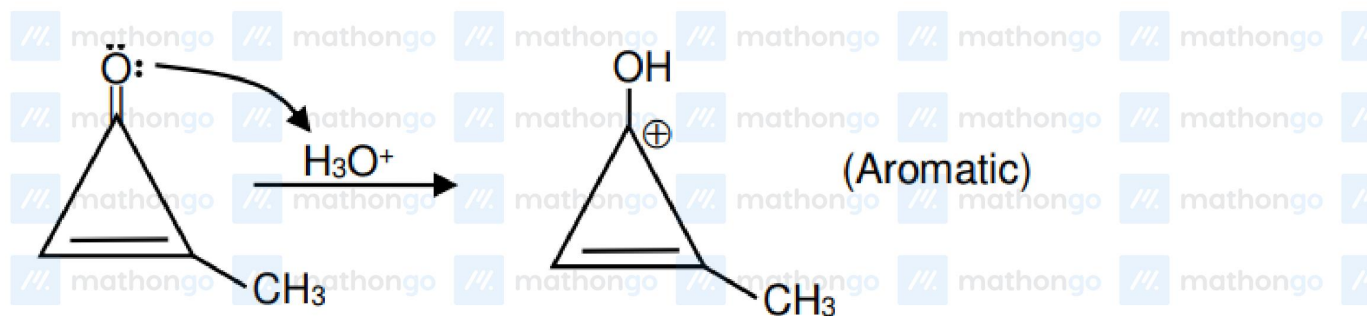
Chemistry



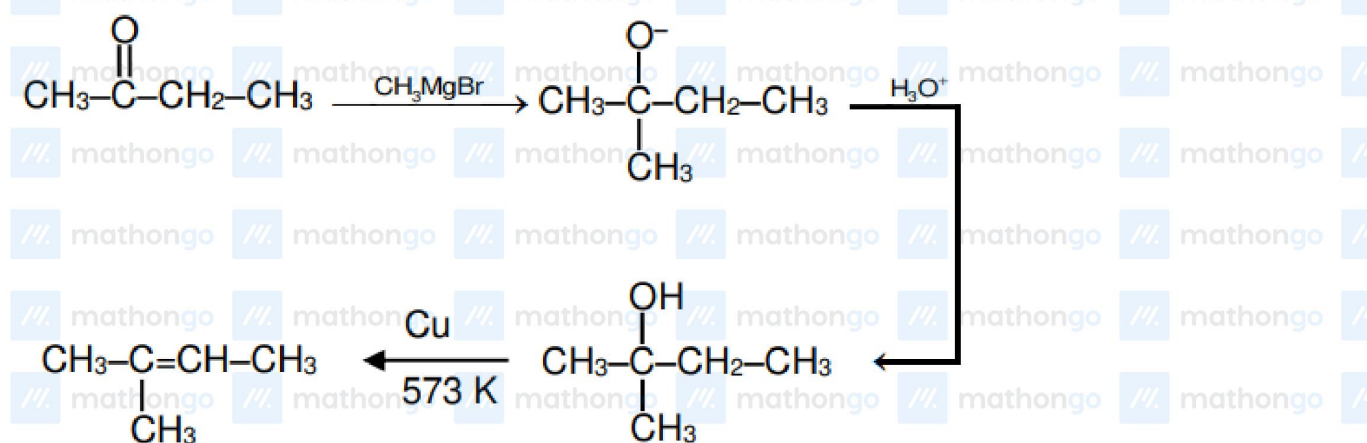
Q9



Q10



Q11



Compound A is CCC(=O)C ($\text{C}_4\text{H}_8\text{O}$)

Percentage carbon in compound

$$A = \left(\frac{12 \times 4}{48 + 16 + 8} \times 100 \right) = 66.67$$